

STM32WL55 Satellite Radio Subsystem

Technical Root Cause Analysis, RF Front-End Architecture & Engineering Fixes

Target Board: STM32WL55JC (Cortex-M0+ Radio Core)	Downlink Frequency: 437.375 MHz
Output Target: ~25.0 dBm (ADL5324 3.3V PA)	Status: Verified Active & Operational
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1. Executive Summary

During flight firmware bring-up on the STM32WL55 Cortex-M0+ radio coprocessor, the RF transmission output measured an anomalous **3.8 dBm** despite the presence of a 3.3V external Power Amplifier (ADL5324) designed for ~25 dBm output. In addition, CW Morse carrier keying suffered from severe tone distortion and startup clipping, PA8 was being needlessly cycled despite power rail 4 being permanently energized in hardware, and GMSK burst packets were transmitting without sequential packet counters.

Through systematic firmware and hardware-level root cause investigation, the RF switch control logic, internal PA routing, oscillator standby state, and packet pipeline were corrected. Measured transmission power immediately increased from **3.8 dBm to ~25.0 dBm (316 mW)**, CW keying envelope was perfected, and dynamic packet numbering was successfully enabled.

2. Comprehensive Technical Root Cause Analysis

2.1 Issue 1: RF Switch Inversion & Dual-Active Conflict (PC4 & PC5)

Previous Code Behavior: The legacy driver treated PC4 as a chip enable (*VEN*) and PC5 as a direction pin (*VCTL*). Consequently, both PC4 and PC5 were driven HIGH (1, 1) during transmission.

Physical Failure Mechanism: The hardware RF switch (SKY13373 / SP3T) uses discrete front-end control lines: **PC4 = FE_CTRL1 (RX path)** and **PC5 = FE_CTRL2 (TX path)**. Driving both lines HIGH simultaneously caused the switch to attempt to engage both RX and TX simultaneously or enter an illegal high-isolation state. Transmit RF power was attenuated by **20 to 30 dB**, with only weak capacitive cross-coupling reaching the antenna.

Engineering Fix: Inverted and separated control states: In TX mode, **PC4 (FE_CTRL1) = 0** (isolating RX) and **PC5 (FE_CTRL2) = 1** (opening TX path to antenna). In RX mode, **PC4 = 1** and **PC5 = 0**.

2.2 Issue 2: RF Output Pin Mismatch (RFO_LP vs RFO_HP)

Physical Architecture: The STM32WL55 silicon contains two independent internal power amplifiers with dedicated physical pins: **PB0 (RFO_LP)** rated up to +14 dBm, and **PB1 (RFO_HP)** rated up to +22 dBm. On the board hardware, the external 3.3V amplifier and matching network are routed directly to the **RFO_HP** pin.

Failure Mechanism: Firmware was hardcoded to RFO_LP (+14 dBm). The radio transmitted RF energy on pin PB0, while the RF switch was connected to pin PB1. The only signal reaching the external PA was substrate cross-talk (~-15 dBm to -10 dBm) between silicon pads PB0 and PB1. Amplified by the external PA, this resulted in the measured **3.8 dBm**.

Engineering Fix: Configured the Sub-GHz radio driver for **RFO_HP at +22 dBm**, opening the current limiter REG_OCP to 160 mA (0x38), enabling REG_TX_CLAMP, and returning RBI_CONF_RFO_HP. Driving +22 dBm directly into the external amplifier produced the target **~25 dBm** output.

2.3 Issue 3: PA8 Conflict & Dedicated PC2 (SO2) PA Enable

Findings: Firmware was configuring and toggling PA8 as `RF_SW_VDD`. On the satellite bus, Power Rail 4 (3.3V) is already permanently energized in hardware, rendering PA8 control redundant and introducing unnecessary bus transitions.

Fix: Excised all PA8 references. Dedicated **PC2 (SO2 / AMP_3V3_EN_PIN)** exclusively to the 3.3V Power Amplifier: driven HIGH (1) only during active transmission and pulled LOW (0) during RX and Standby to eliminate quiescent current.

2.4 Issue 4: CW Morse TCXO Shutdown Bug (STDBY_RC vs STDBY_XOSC)

Failure Mechanism: In `Satellite_CW_CarrierOff()`, the radio was commanded into `STDBY_RC` between Morse dots and dashes. This powered down the 32 MHz TCXO and PLL synthesizer. Restarting the TCXO and locking the PLL required 5 to 10 ms before every dot and dash. For an 80 ms Morse dit, this restart latency severely clipped the leading edge of each tone.

Fix: Switched carrier off to `STDBY_XOSC`. This immediately cuts RF carrier while keeping the 32 MHz TCXO running continuously, enabling instantaneous (<15 μ s) carrier keying with sharp, full-power pulse envelopes.

2.5 Issue 5: Lead-In Settle & Trailing Ramp-Down Guard Delays

Lead-In Delay: Added a ~200 μ s settling loop in `RBI_ConfigRFSwitch()` after asserting PC2 and PC5 before RF energy is driven, allowing external PA bias decoupling capacitors to fully stabilize.

Trailing Guard: Added a 2 ms hold delay in `Satellite_Send_Packet_Timeout()` after `Radio_Send_And_Wait()` before deasserting PC2. Previously, the PA was disabled the exact microsecond the TX Done IRQ fired, cutting off the final CRC flags during SX1262 internal ramp-down.

2.6 Issue 6: Dynamic GMSK Burst Packet Numbering

Implementation: Added a lifetime packet counter `s_total_packets_sent`. Inside the transmission loop of `Satellite_Run_GMSK_Burst_Session_Ex()`, each packet payload is dynamically formatted with `#<number>` (e.g., *"Namaste everyone, Testing GMSK signal #1"*), encoded into AX.25 UI frames, scrambled with G3RUH, and transmitted. The cycle completion status telemetry packet also reports the total packet counter.

3. Firmware Logic & Parameter Comparison Table

Parameter	Old Implementation	Corrected Implementation	Engineering Impact
PA8 Supply Control	Toggled on every TX/RX	Completely Removed	Hardware power rail 4 is permanently ON; eliminates delay
PC2 (3.3V PA Enable)	Grouped / Indeterminate	TX=1, RX=0, OFF=0	Powers 3.3V external PA (ADL5324) during TX only
PC4 (FE_CTRL1) in TX	HIGH (1) [WRONG]	LOW (0) [CORRECT]	Completely isolates RX LNA during transmit
PC5 (FE_CTRL2) in TX	HIGH (1)	HIGH (1)	Opens low-loss path from transmitter to antenna
PC4 (FE_CTRL1) in RX	HIGH (1)	HIGH (1)	Directs antenna incoming signal to RX LNA
PC5 (FE_CTRL2) in RX	LOW (0)	LOW (0)	Isolates transmit path during reception
Radio Silicon PA Output	RFO_LP (PB0 @ 14 dBm)	RFO_HP (PB1 @ 22 dBm)	Eliminates silicon cross-talk; routes directly to external PA
CW Inter-Element Standby	STDBY_RC (TCXO killed)	STDBY_XOSC (TCXO ON)	Zero TCXO wakeup latency; perfect CW tone keying envelope
PA Turn-On Settle Delay	None	~200 μs lead-in delay	Allows PA bias network to stabilize before RF drive
Post-TX Guard Delay	0 ms (immediate cutoff)	2 ms guard delay	Prevents truncating packet trailing flags / CRC bits
GMSK Burst Numbering	Static pre-encoded string	Dynamic (#1, #2, ...)	Provides exact packet tracking and loss accounting at GS
Measured RF Output	3.8 dBm (Weak)	~25.0 dBm (Strong)	SUCCESS: Full 300+ mW amplification achieved

4. Verification & Operational Status

1. **RF Power Output:** Measured power increased from 3.8 dBm to **~25 dBm** on spectrum analyzer / power meter, confirming full drive into the 3.3V external PA.
2. **Flashing Operation:** satellite.bin was successfully verified and flashed to address 0x08032000 using STM32CubeProgrammer CLI with verified readback.
3. **Packet Continuity:** GMSK AX.25 packets transmit continuously with dynamic sequential numbering logged over USART1 (115200 baud).